Cricket Player Performance Prediction Using Machine Learning

Infosys Springboard

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Abstract

Cricket is a data-driven sport where millions of data points are generated every season through ball-by-ball match records. Extracting meaningful insights from this large volume of data is a challenging task. Traditional statistics such as batting average or strike rate do not fully capture the impact of player form, venue conditions, and opposition strength.

This project proposes a Machine Learning based system to predict the performance of cricket players in upcoming matches. Two separate predictive models are developed: one for predicting the runs scored by a batsman and another for predicting the wickets taken by a bowler. Historical IPL data from 2008 to 2025 is used for training and evaluation.

Advanced feature engineering techniques such as rolling averages, career statistics, venue-based performance, and opponent-based performance are applied. Models including XGBoost Regressor and Random Forest Regressor are trained and evaluated using standard regression metrics.

An interactive Streamlit dashboard is developed to allow users to select players, teams, and venues, and instantly obtain predictions along with performance visualizations and explainable AI outputs using SHAP.

The proposed system demonstrates the effectiveness of Machine Learning in sports analytics and provides a practical decision-support tool for analysts, coaches, and cricket enthusiasts.

CHAPTER 1

INTRODUCTION

1.1 Background

Cricket has evolved from a purely skill-based sport to a highly analytical discipline. Teams now rely heavily on data analysis for player selection, strategy formulation, and match preparation. The Indian Premier League (IPL), being one of the most competitive cricket leagues in the world, generates massive amounts of data for every ball bowled.

Analyzing this data manually is impractical. Machine Learning offers automated methods to discover hidden patterns and relationships that can be used to make accurate predictions.

1.2 Motivation

Predicting how a player will perform in the next match is valuable for:

- Team management
- Coaches
- Fantasy cricket platforms
- Fans and analysts

Accurate predictions help in optimizing team combinations and planning match strategies.

1.3 Problem Definition

The problem is to design a system that predicts:

- Runs scored by a batsman in an upcoming match
- Wickets taken by a bowler in an upcoming match

using historical performance and contextual features.

1.4 Objectives

- Collect and preprocess IPL datasets
- Engineer meaningful features
- Train predictive models
- Evaluate model performance
- Build an interactive dashboard

CHAPTER 2

LITERATURE REVIEW

Several studies have explored Machine Learning applications in sports analytics.

Researchers have used Decision Trees, Random Forests, Support Vector Machines, and Neural Networks to predict match outcomes and player performance. Studies show that ensemble models such as Random Forest and Gradient Boosting provide better accuracy than simple linear models.

However, many existing works focus only on match outcome prediction. Player-level performance prediction with explainability is still limited. This project addresses this gap.

CHAPTER 3

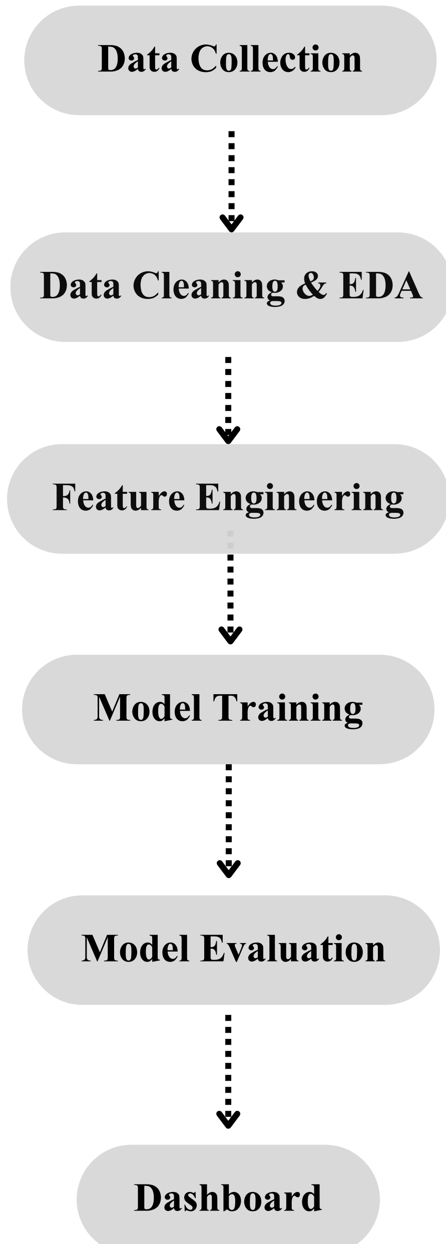
SYSTEM ANALYSIS & TECH STACK

3.1 Introduction

The overall system design and working methodology of the Cricket Player Performance Prediction System. The system integrates data engineering, machine learning modeling, explainable artificial intelligence, and an interactive web dashboard to provide accurate and interpretable predictions of cricket player performance.

The complete workflow starts from raw historical IPL datasets and ends with a Streamlit-based dashboard where users can select match conditions and obtain predictions for batsmen and bowlers along with visual explanations. The system is modular in nature, allowing each component such as data preprocessing, feature engineering, model training, and deployment to operate independently while being seamlessly integrated into a single pipeline.

3.2 System Architecture



3.3 Technology Stack

- Language: Python
- Libraries: Pandas, NumPy, Scikit-learn
- Models: XGBoost, RandomForest
- Explainability: SHAP
- Deployment: Streamlit Framework
- Version Control: Git

3.4 Dataset Description

Source: Kaggle
IPL Dataset (2008–2025)

Dataset	Rows	Columns
ball_by_ball_data	278,205	30
ipl_matches_data	1,169	23
players-data-updated	772	6
team_aliases	46	3
teams_data	16	2

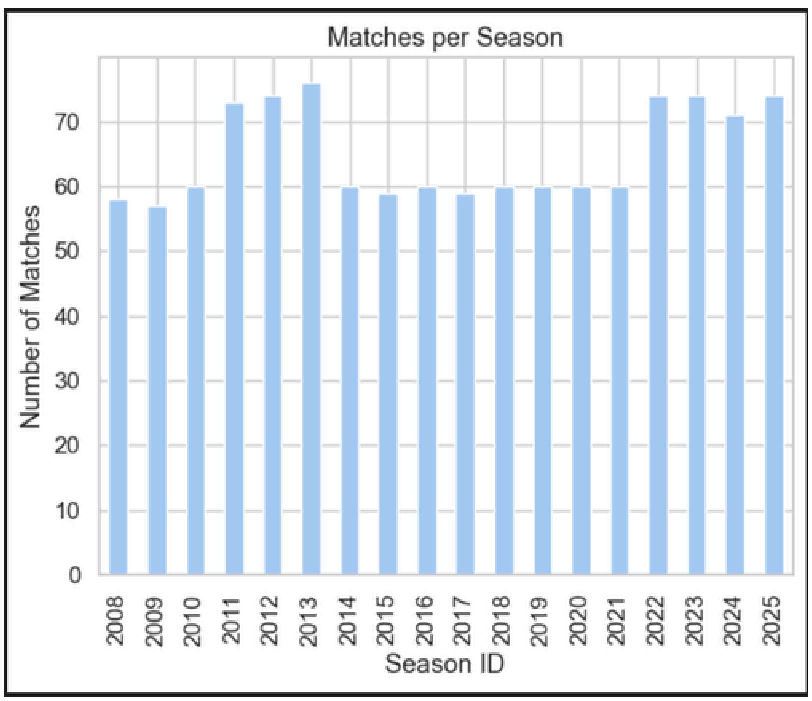
3.5 Data Cleaning, preprocessing

- Removal of irrelevant columns
- Handling missing values
- Removal of duplicates
- Standardization of team names
- Encoding categorical features

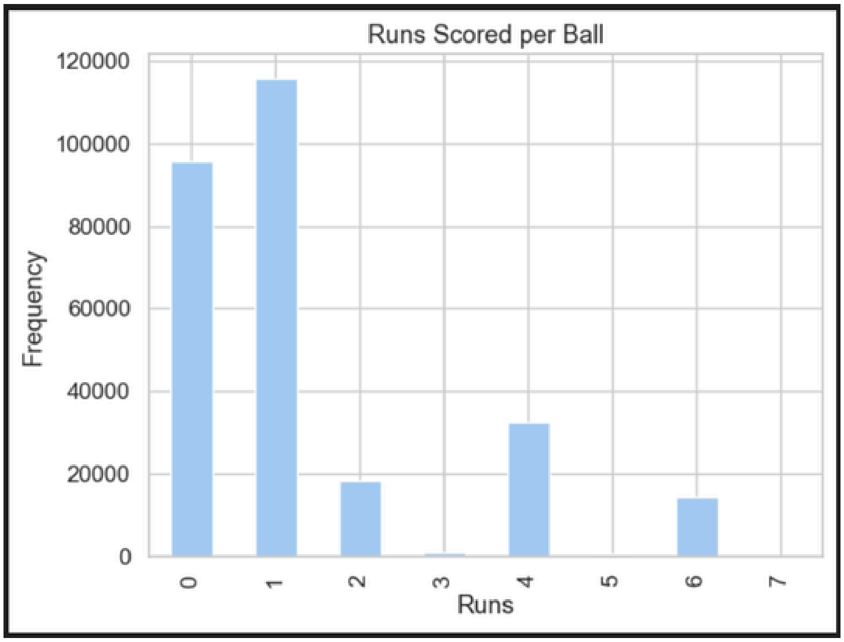
3.6 Exploratory Data Analysis (EDA)

- EDA is the critical process of investigating, cleaning, and summarizing datasets using visualization and statistical methods to uncover patterns, detect anomalies, and test assumptions

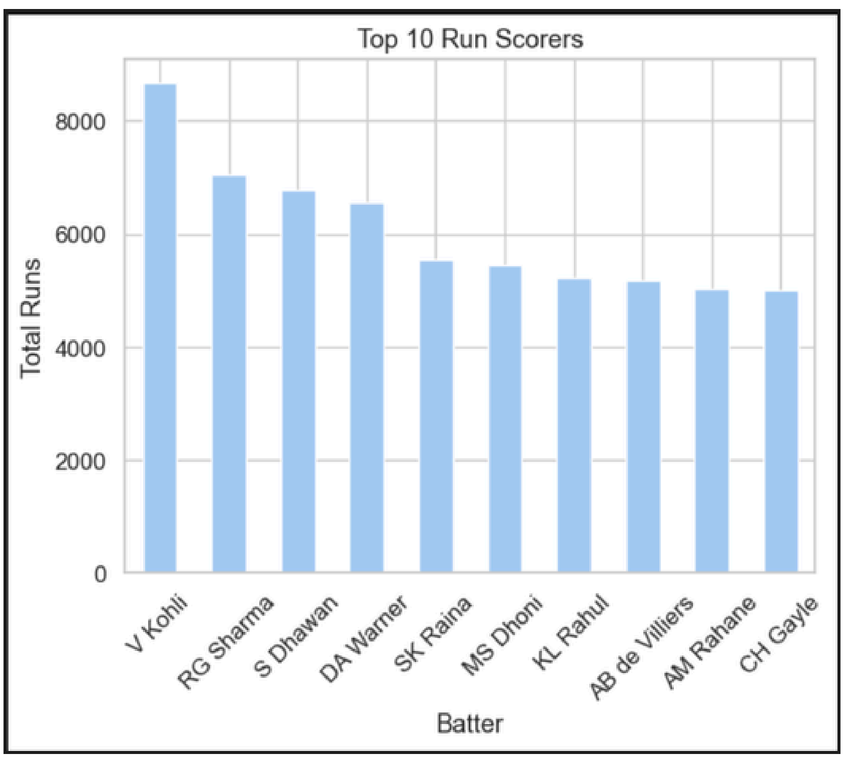
3.6.1 MATCHES PER SEASON



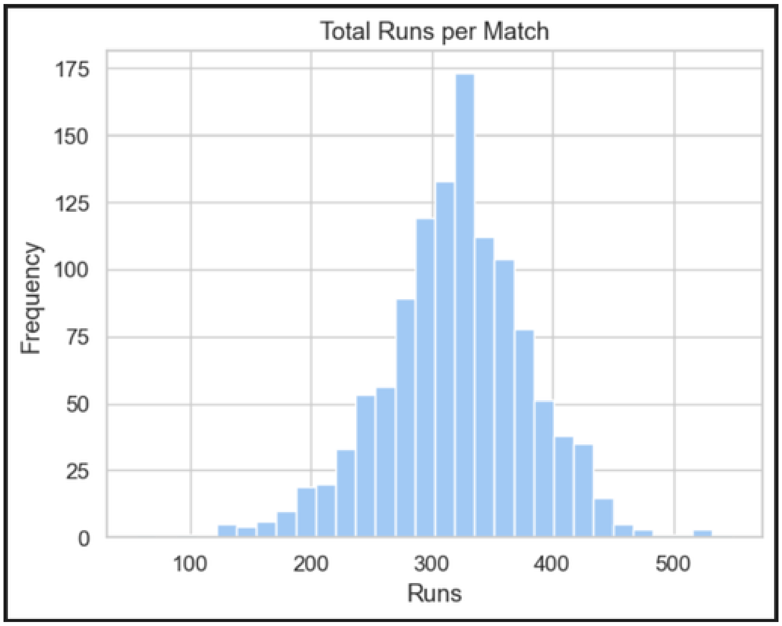
3.6.2 AVERAGE RUNS SCORED PER BALL



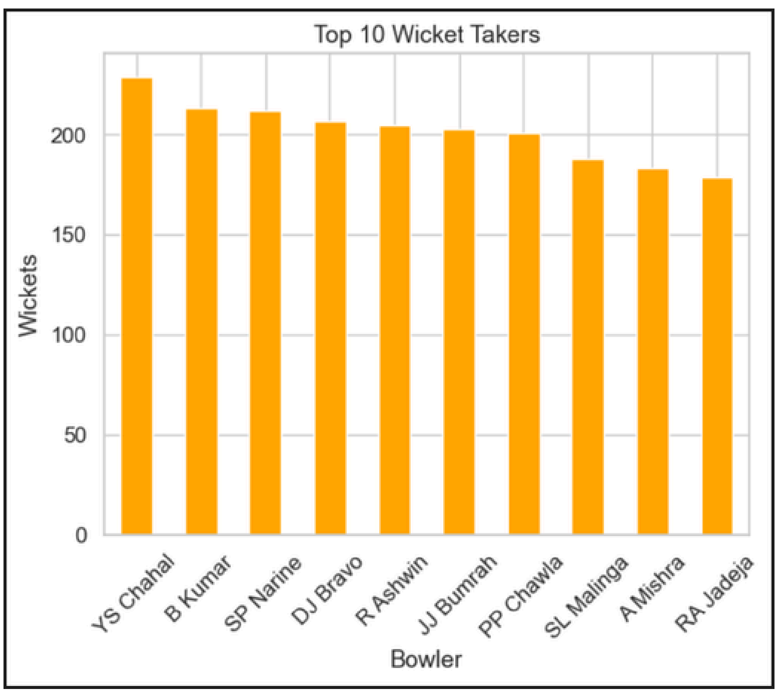
3.6.3 TOP 10 RUN SCORERS



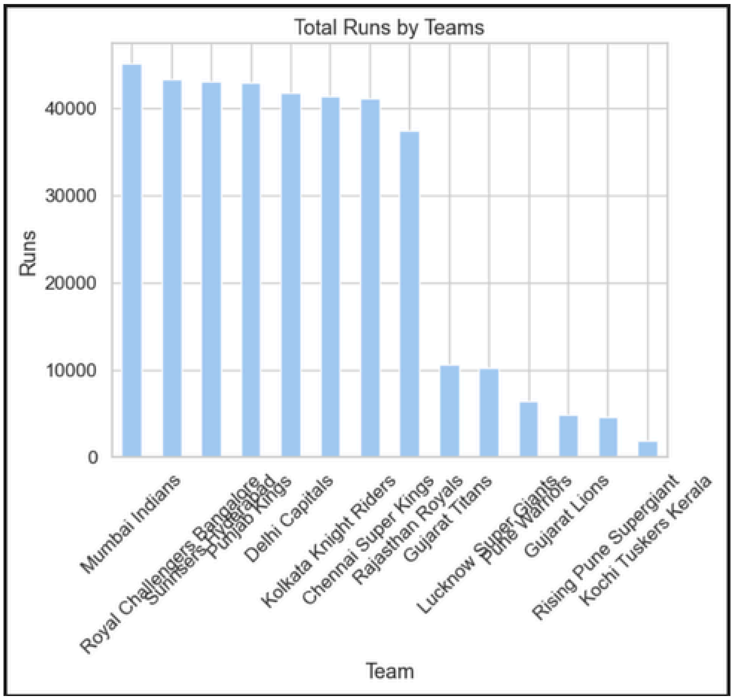
3.6.4 TOTAL RUNS PER MATCH



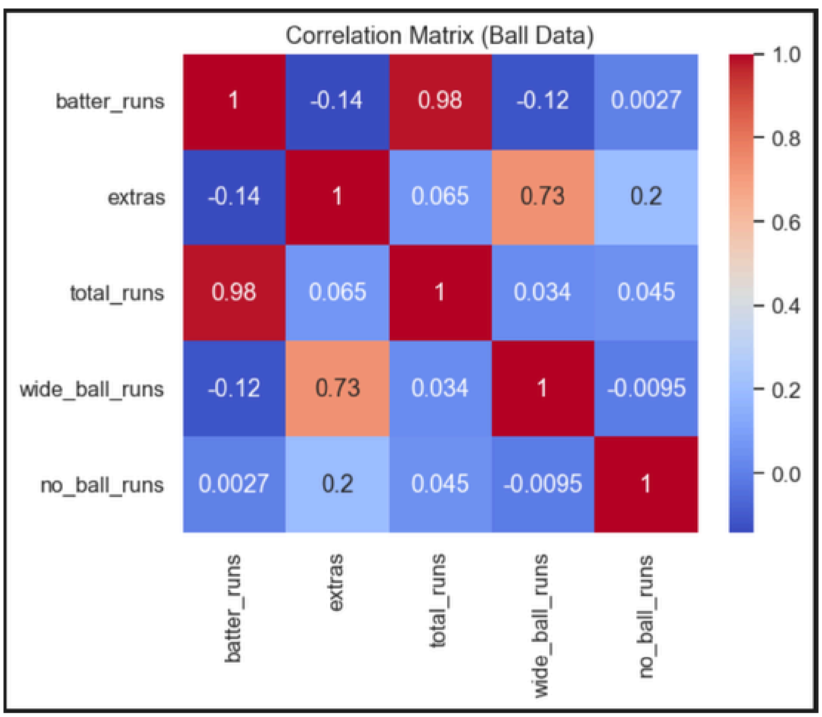
3.6.5 TOP 10 WICKET TAKERS



3.6.6 TOTAL RUNS BY TEAMS



3.6.7 Correlation Matrix (Ball Data)



CHAPTER 4

FEATURE ENGINEERING

4.1 Introduction

Feature engineering is a crucial part of the system as it transforms raw cricket data into meaningful numerical features that represent player ability and recent form.

Examples of engineered features include:

- Player form features (last 5 and last 10 match performance)
- Career statistics (career average, strike rate, total matches)
- Venue-based averages
- Opposition-based averages
- Frequency encoded categorical variables

Two final feature sets are generated:

- Batsman Feature Set
- Bowler Feature Set

These features form the input to the machine learning models after encoding as we did:

- Label Encoding
- Frequency Encoding
- One-hot encoding

CHAPTER 5

MODEL DEVELOPMENT & EVALUATION

5.1 Introduction

To develop a robust prediction system, multiple machine learning regression models were trained separately for batsmen and bowlers. Training multiple models allows comparative evaluation and ensures that the most accurate model is selected for deployment.

The following regression algorithms were implemented:

- Random Forest Regressor
- XGBoost Regressor
- LightGBM Regressor

Each model was trained using engineered features derived from historical IPL match data.
To evaluate the performance of the machine learning models fairly, the dataset was divided into:

- Training Set – 80%
- Testing Set – 20%

◆ Why 80–20 Split?

- 80% data is used to train the model
- 20% data is used to test model performance
- Prevents overfitting
- Ensures unbiased evaluation

5.2 Batsman Performance Models

For predicting the number of runs a batsman will score in the next match, three models were trained:

- Random Forest Regressor
- XGBoost Regressor
- LightGBM Regressor

After initial training, hyperparameter tuning was performed using Grid Search and Randomized Search techniques to optimize model parameters such as:

- Number of trees
- Maximum depth
- Learning rate
- Subsample ratio
- Column sampling rate

The tuned models were evaluated using MAE, RMSE, and R^2 score. Among all three models, XGBoost Regressor achieved the best performance, producing the lowest error values and highest R^2 score.

Therefore, XGBoost was selected as the final model for batsman run prediction.

5.3 Bowler Performance Models

For predicting the number of wickets a bowler will take in the next match, the same three models were trained:

- Random Forest Regressor
- XGBoost Regressor
- LightGBM Regressor

Hyperparameter tuning was again applied to each model to improve accuracy.
After comparison of evaluation metrics, Random Forest Regressor demonstrated superior performance compared to others. Hence, Random Forest was selected as the final model for bowler wicket prediction.

5.4 Model Comparison Strategy

The selection of final models was based on:

- Lowest Mean Absolute Error (MAE)
- Lowest Root Mean Squared Error (RMSE)
- Highest R^2 Score
- Stability of predictions

This systematic comparison ensured that the chosen models provide reliable and consistent predictions.

Task	Selected Model
Batsman Run Prediction	XGBoost Regressor: XGB MAE: 9.045175558281114 BEST XGB RMSE: 12.169984300340385 BEST XGB R2: 0.7053843337317794
Bowler Wicket Prediction	Random Forest Regressor: RF MAE: 0.8295097084031978 Tuned RF RMSE: 1.05548181481273 Tuned RF R2: 0.04568238885329012

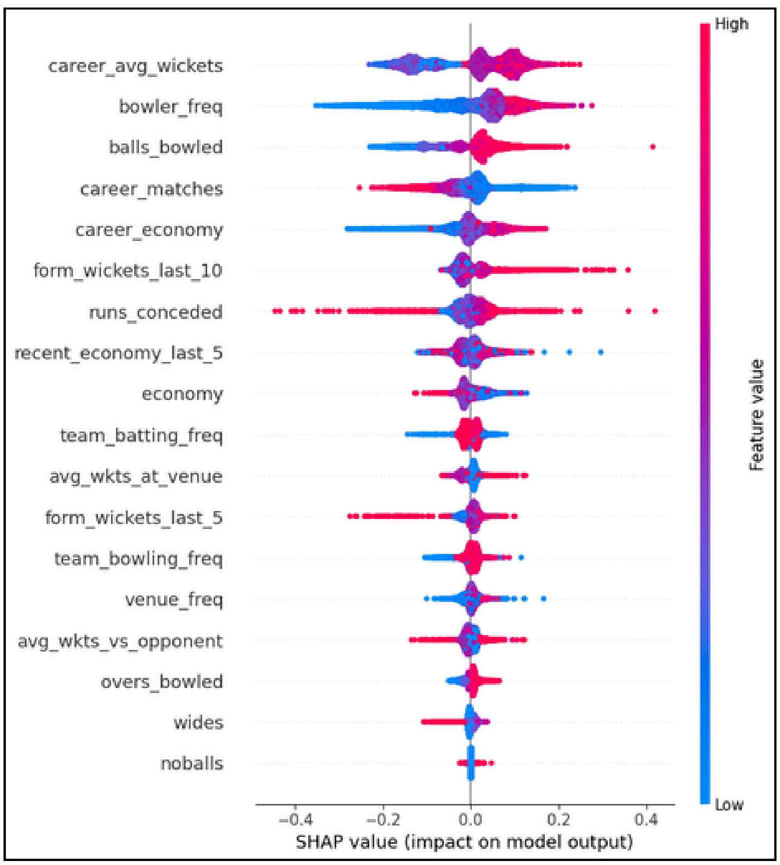
5.6 Explainability Layer (SHAP)

To improve transparency and trust, SHAP (SHapley Additive Explanations) is integrated.

SHAP explains:

- Which features influenced a prediction
- The magnitude of each feature's impact
- Positive or negative contribution

This allows users to understand why a certain prediction was generated.



CHAPTER 6

DASHBOARD

6.1 Introduction

To make the prediction system usable by non-technical users, a web-based interactive dashboard was developed using Streamlit. The dashboard allows users to select match inputs such as batsman, bowler, teams, and venue, and instantly view predicted performances along with visual insights.

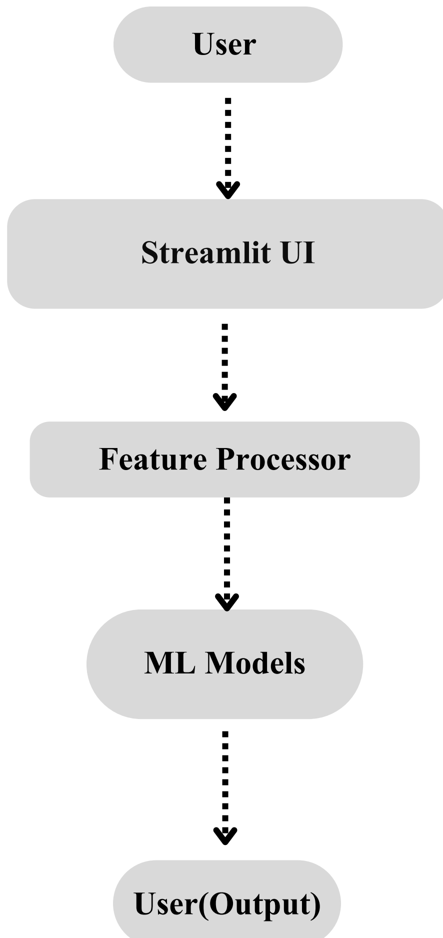
The dashboard acts as the interface between trained machine learning models and end users.

6.2 Objectives of the Dashboard

The main objectives of the dashboard are:

- Provide a simple and interactive interface for prediction
- Accept real-time match inputs
- Display predicted runs and wickets
- Visualize past performance trends
- Explain predictions using SHAP interpretability

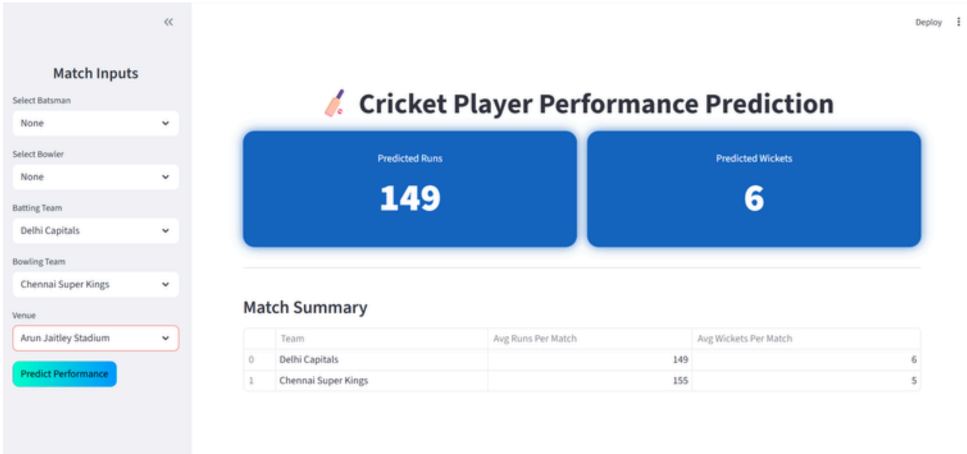
6.3 Dashboard Architecture



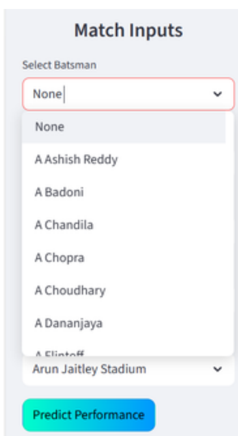
6.4 Input Interface

Users can select the following inputs:

- Batsman (optional)
- Bowler (optional)
- Batting Team
- Bowling Team
- Venue



Dropdown menus automatically load player, team, and venue names from datasets.



6.5 Prediction Logic

Two modes are supported:

1. Player-Based Prediction
When batsman and bowler are selected:
- XGBoost model predicts runs
- Random Forest model predicts wickets

2. Team-Based Prediction

When player is not selected:

- Team historical averages are used
- Provides reasonable estimates for match-level predictions

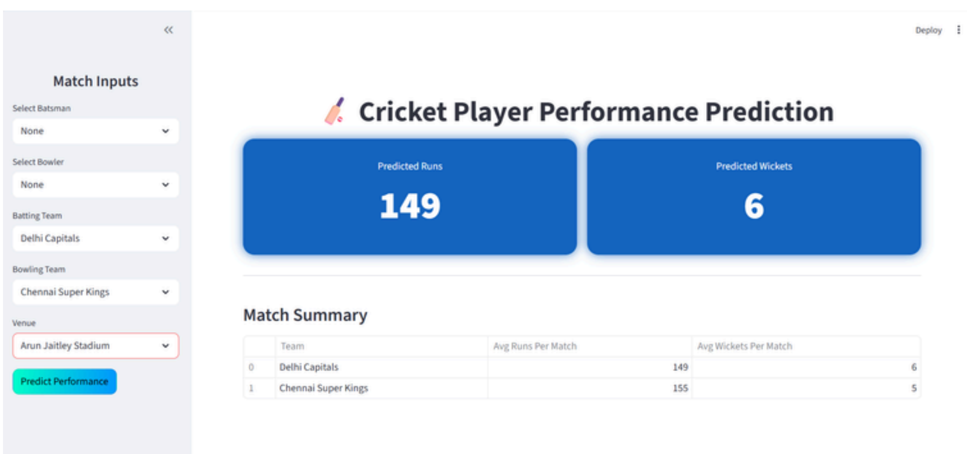
This ensures the system remains functional even when player details are unavailable.

6.6 Output Display

The dashboard displays:

- Predicted Runs
- Predicted Wickets

Results are shown using visually highlighted cards for easy understanding.

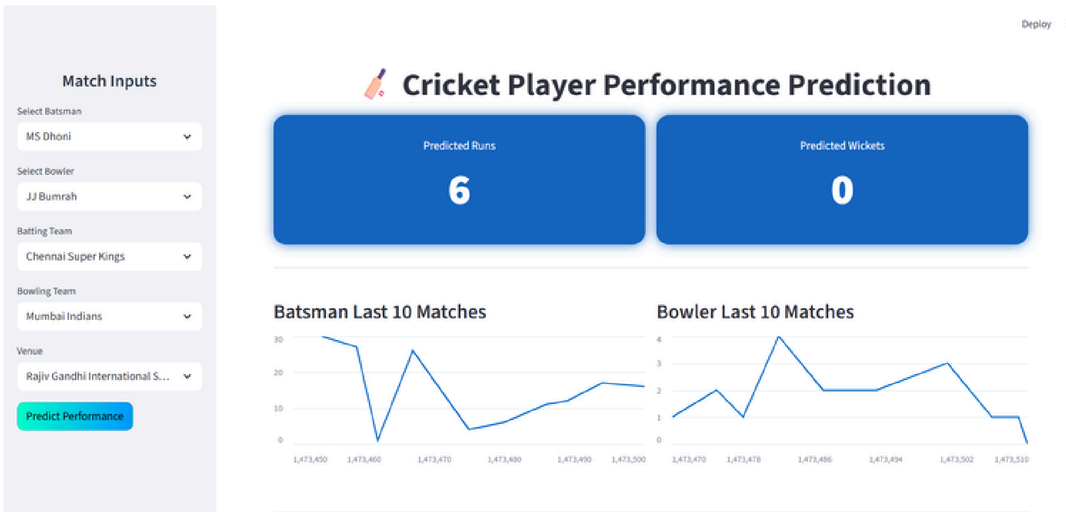


For selected players:

- Line chart showing batsman runs in last 10 matches

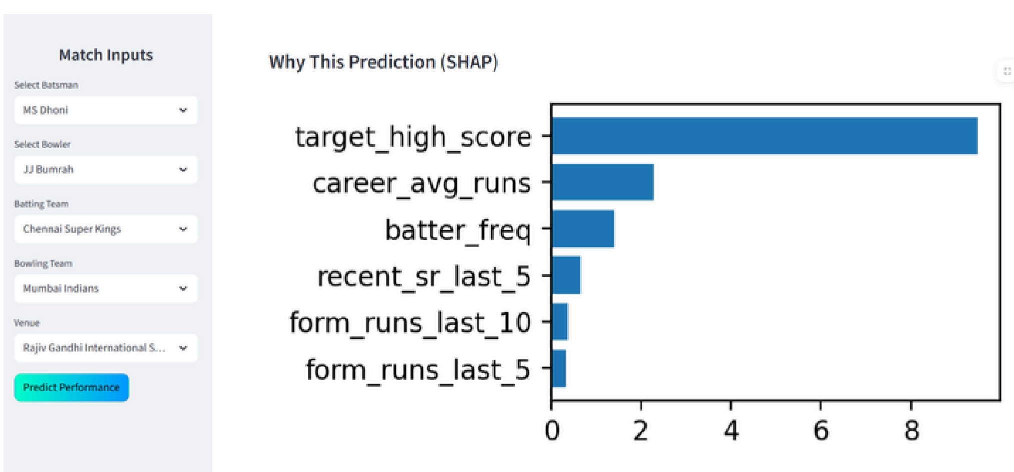
- Line chart showing bowler wickets in last 10 matches

These charts help users understand recent form.



SHAP (SHapley Additive exPlanations) is integrated to explain predictions.

- The dashboard displays a small bar chart showing the most influential features affecting the batsman prediction.
- This improves transparency and trust in the system.



6.7 Advantages of the Dashboard

- User-friendly
- Fast predictions
- Interactive visuals
- Explainable AI
- Scalable

CHAPTER 7

CONCLUSION

This project successfully designed and implemented an AI-based Cricket Player Performance Prediction System that predicts a batsman's expected runs and a bowler's expected wickets for upcoming IPL matches using historical data and machine learning techniques. Large-scale IPL datasets covering multiple seasons were collected, cleaned, and analyzed. Extensive feature engineering was performed to capture player career statistics, recent form, opponent-specific performance, and venue-specific trends. These features significantly improved the predictive capability of the models.

Multiple machine learning algorithms including Random Forest, XGBoost, and LightGBM were trained separately for batsmen and bowlers. After comprehensive evaluation and hyperparameter tuning, XGBoost was selected for batsman run prediction, while Random Forest was selected for bowler wicket prediction, based on superior performance across MAE, RMSE, and R^2 metrics.

To make the system practical and accessible, an interactive Streamlit dashboard was developed. The dashboard enables users to input match details, view predictions instantly, analyze recent player form through visualizations, and understand model decisions using SHAP explainability. This combination of prediction and interpretability increases user trust and usability.

Overall, the system demonstrates how machine learning and data-driven techniques can assist in sports analytics, providing valuable insights for analysts, fantasy cricket players, coaches, and fans. The project fulfills all proposed objectives and establishes a strong foundation for future enhancements such as real-time data integration, pitch and weather analysis, and advanced deep learning models.